Assignment - 2

1. Explain SaaS with an example

- SaaS is a software distribution model where applications are hosted by a provider and accessed via Internet using a browser.
- The key features of SaaS are :
  - no installation
  - no maintenance
  - no infrastructure management required.

- The example of a SaaS platform is :
Google Workspace, users can access Gmail, Docs, Sheets & Drive directly online without installing software locally. The other examples are Microsoft 365, Salesforce.

2. What are the challenges faced by SaaS?

i) Data Security & Privacy
- Risk of breaches, unauthorized access, & data loss. It requires encryption & decryption compliance.

ii) Internet Dependency
- Downtime or poor connectivity disrupt access.

iii) Limited Customization
- The multi-tenancy restricts UI / feature modifications.

iv) Integration Complexity
- It's difficult to connect SaaS with legacy systems, requires APIs or middleware.

v) Vender lock-in
- Migration between providers is costly & risky...

vi) Performance & Reliability
- Shared infrastructure may cause latency or service interruption

vii) Compliance and Legal Risks
- Data stored across borders may violate residency laws.

3. Justify "SaaS Integration is "Hard"

→ i) Different API standards
- The REST vs SOAP, version mismatches.

ii) Real-time sync issue
- Data Consistency across multiple apps is difficult.

iii) Error Handling Complexity
- logging & debugging across platform is challenging.

iv) Legacy System Integration
- The old systems may not support modern APIs.

v) Frequent Updates
- SaaS vendors push changes that may break integrations.

- The integration requires technical expertise, middleware or iPaaS solutions, making it complex & time consuming.

4. Explain SaaS Integration Services

→ Businesses use multiple SaaS tools that don't communicate automatically. Integration ensures smooth workflows.
- Types of services:

i) APIs
- allows apps to exchange data.

ii) Middleware
- connect multiple SaaS apps.

iii) iPaaS
- Cloud based tools for enterprise-level integration, real time syncing & workflow orchestration.

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5. Explain SaaS Integration of Products & Platforms.

→ Product Integration

i) CRM. (Salesforce)
- Integrated with marketing tools, ERP, email.

ii) ERP (SAP, Netsuite)
- connected with HRM, finance, logistics.

iii) HRM (Workday, BambooHR)
- linked with payroll, recruitment tools.

iv) Marketing (Hubspot, Marketo)
- Integrated with CRM, analytics, email

v) E-commerce (Shopify)
- connected with inventory, shipping, analytics.

- The pros of product integration are centralized data, improved customer experience, streamlined operations.

→ Platform Integration.

i) AWS.
- hosting SaaS products, serverless compute, AI

ii) Microsoft Azure
- enterprise services, Active Directory for identity management

iii) Google Cloud

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- Big data, APIs, ML/AI

ii) Heroku
- PaaS for deploying SaaS apps.

iii) use cases of platform integration are hosting apps, using APIs, CI/CD integration with github.

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